

# **Gimbal Mount Assembly**

## Manufacturing, Assembly, Integration and Test Plan

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# **1. Introduction**

## **1.1. Scope**

The purpose of the Manufacturing, Assembly, Integration and Test (MAIT) Plan is to describe the MAIT process and demonstrate together with the verification plan how the requirements are verified by inspection and test. It contains the overall Manufacturing, AIT activities, suggestions for associated verification tools (GSE - Ground Support Equipment) and the involved documentation.

## **1.2. Reference**

[RD1] Gimbal Mount Assembly - Definition File / Version 001

[RD2] Gimbal Mount Assembly - Interface Control Document / Version 001

[RD3] Data Sheet - ARMCO 17-4PH / 2022

[RD4] Instructions for Double Anvil Spherical Bearing Swaging Tool #10 Wide / Pegasus Auto Racing Supplies, Inc. /

[RD5] Bearing Installation and Retention by Swaging or Staking / AIA - NAS0331 / 2013

[RD6] NASA Reference Publication 1228 - Fastener Design Manual / Richard T.Barrett / 1990

## 2. GMA Presentation

As the primary thrust-transmitting interface, the Gimbal Mount Assembly (GMA) connects the engine to the vehicle, transfers engine thrust to the vehicle structure, and allows the engine to gimbal relative to the vehicle structure for thrust-vector control. The design is detailed in the Definition File [RD1] and the Interface Control Document [RD2].

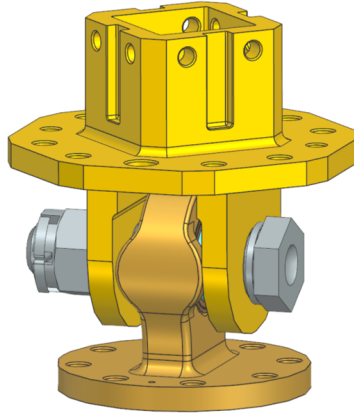


Figure 1: GMA isometric view

## 3. Manufacturing

The manufacturing process shall follow the companies established and proven internal processes from sourcing of the raw material until final manufacturing and subsequent quality related processes. The following figure shows the parts that shall be custom machined, while the rest are of COTS-nature as described in [RD1].

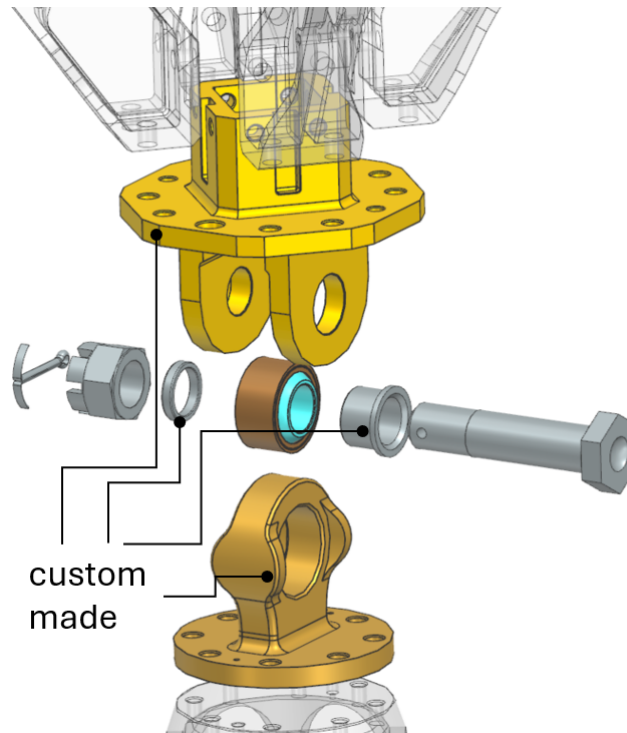


Figure 2: Custom machined parts of the TMA

### 3.1. Materials and Heat Treatment

All custom machined parts of the GMA (Figure 2) are made of martensitic precipitation hardened stainless steel, that combines high strength, excellent corrosion resistance, easy heat treatment and comparable thermal expansion w.r.t Inconel 718 (engine). As one of the main structural parts of the engine, the GMA is treated as a critical application. For this, the material shall be sourced in a precipitation hardened condition. The final material choice is **17-4PH - H900**, while the suffix expresses the heat treatment condition. The figure below shows a general overview of material properties as a function of the precipitation hardened conditions ranging from *H900* to *H1150M*.

| Property                               | Condition   |            |            |            |            |            |           |           |
|----------------------------------------|-------------|------------|------------|------------|------------|------------|-----------|-----------|
|                                        | H900**      | H925       | H1025      | H1075      | H1100      | H1150      | H1150-D   | H1150M    |
| UTS, MPa (ksi)                         | 1379 (200)  | 1310 (190) | 1172 (170) | 1138 (165) | 1034 (150) | 1000 (145) | 965 (140) | 862 (125) |
| 0.2% YS, MPa (ksi)                     | 1276 (185)* | 1207 (175) | 1138 (165) | 1034 (150) | 931 (135)  | 862 (125)  | 758 (110) | 586 (85)  |
| Elongation % in 4D <sub>0</sub>        | 14.0        | 14.0       | 15.0       | 16.0       | 17.0       | 19.0       | 20        | 22.0      |
| Reduction of Area %                    | 50.0        | 54.0       | 56.0       | 58.0       | 58.0       | 60.0       | 60.0      | 68.0      |
| Brinell Hardness                       | 420         | 409        | 352        | 341        | 332        | 311        | 302       | 277       |
| Rockwell Hardness                      | C44         | C42        | C38        | C36        | C35        | C33        | C31       | C27       |
| Impact, Charpy<br>*V-Notch, J (ft•lbs) | 20 (15)     | 34 (25)    | 47 (35)    | 54 (40)    | 61 (45)    | 68 (50)    | 108 (80)  | 135 (100) |

\*0.2% Compressive YS – 1227 MPa (178 ksi)

\*\*In applications requiring greater impact toughness, aging for four hours will develop typical properties: UTS – 1351 MPa (196 ksi).

0.2% YS – 1248 MPa (181 ksi). Elong in 50 mm (2") – 14%. Reduction of Area – 52%. Rockwell Hardness C43.

Impact Charpy – 27 J (20 ft•lbs)

Figure 3: 17-4PH, Typical Mechanical Properties - Bar

In case of sourcing issues of the chosen heat treatment condition *H900*, the alternative condition *H1025* can be taken for all parts consistently.

### 3.2. Machining

The GMA Lug Head and Clevis Head are recommended to be manufactured using a 5-axis CNC machine to achieve the required precision and produce the complex geometries specified in the manufacturing drawings.

The Spacer and Bushing shown in Figure 2 may be manufactured by precision turning, as these components determine the radial clearance and the axial positioning accuracy of the adjacent parts.

The final manufacturing strategy shall be determined by the manufacturer, provided that all requirements specified in the manufacturing drawings are met.

## 4. Assembly

This chapter describes the required tools and procedures to assemble the GMA.

### 4.1. Required Tools

The tools required mount the GMA are listed below:

- Hydraulic press to position the Spherical Bearing *MS14103-10* with the GMA Lug Head with a required force of 21.375 lbs +/- 3,000 (95kN +/-13kN) [RD4]
- Double Anvil Spherical bearing Swaging Tool for Spherical Bearing *MS14103-10* in accordance aligned to [RD5]
- Troque Wrench for Bolt NAS6710D29 and Nut MS9358-016
- Torque Wrench for ISO4017-M6x16/25-A4-70 hexagon head screws and ISO4032-M6-A4-70 hexagon regular nuts
- Pliers to bend the Cotter Pin in place

## 4.2. Assembly Sequence

The sequence of the assembly is described in this chapter. The subsequent description shall be accompanied during practical execution along side the dedicated 3D data (CAD-ID: 126703/01).

The assembly is divided in two main steps (Figure 4).

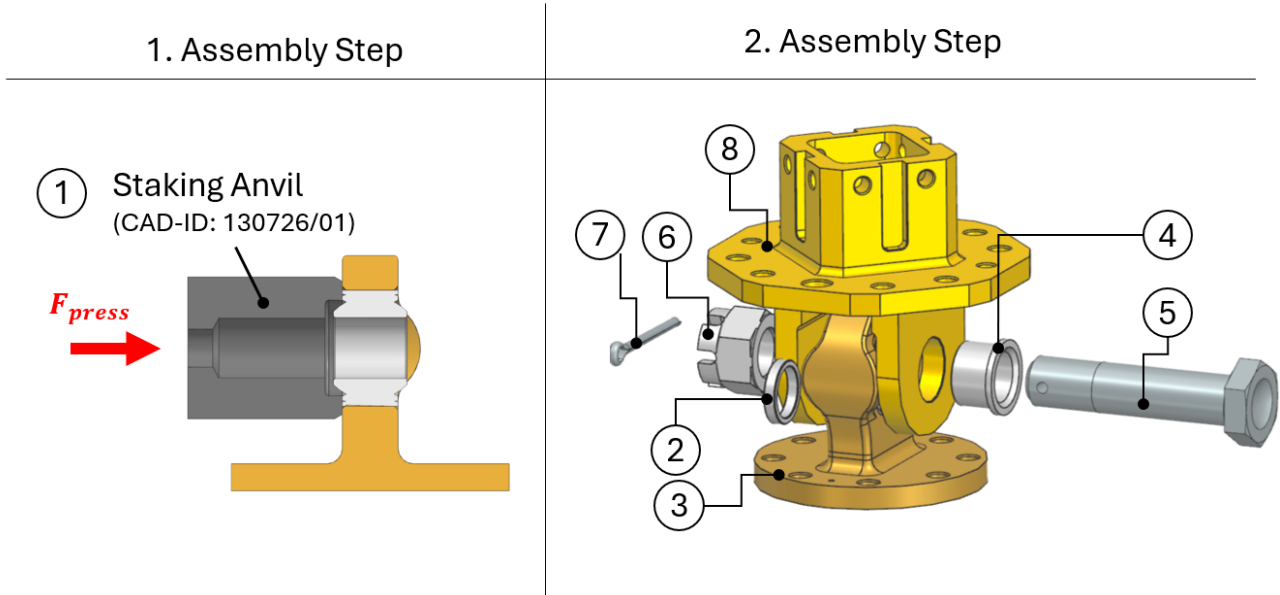


Figure 4: GMA - Assembly steps

In the **first step** (Figure 4, left), the Spherical Bearing *MS14103-10* is inserted into the Clevis Lug Head. Due to the interference fit between the Inner Race of the Spherical Bearing and the Lug Head from  $-0.0055\text{ mm}$  to  $+0.0202\text{ mm}$ , a mounting force will be required. This force shall be centrally introduced by a hydraulic press through a customized Anvil (CAD-ID: 2020/01). The purpose of this customized GSE tool is to position the bearing so that it's width is aligned with the width of the GMA Lug Head. In addition, the tool allows the mounting force to be transmitted through the outer race of the Spherical Bearing only and thus protect the PTFE-fabric liner from being damaged or worn prior engine operation. The anvil swaging of the bearing over housing is then executed following *Method 100* [RD5]. Further instruction details are to be extracted from the manual [RD4] or in chapter 8.1.

The **second step** (Figure 4, right) shows the sequence of the remaining part to the full assembly. While the GMA Lug Head and the Spherical Bearing is united through swaging as mentioned above and detailed in [RD4] and [RD5], this subassembly is then connected to the GMA Clevis Head. The remaining parts exhibit a clearance fit, so that no specific precautions and preparations are needed to make the assembly (and later disassembly) possible under ambient temperature. Following the numbering sequence in the image from two to eight, the Spacer (2) and the GMA Lug Head subassembly (3) are coaxially centered within the GMA Clevis Head holes (8). It is important to note, that the GMA Clevis Head Ears are designed asymmetrically for functional reasons. Hence, the Spacer (2) is positioned at the Clevis inner side between the smallest bore of the Clevis and the Inner Race of the Lug-subassembly. The Bushing (4) is then axially inserted through the bigger bore from outside on the opposite side of the Clevis until it is axially in contact with the inner race of the Spherical Bearing. Once the Spacer (2), the Lug-subassembly (3) and the Bushing (4) are positioned in place, the Bolt *NAS67103D29* (5) is pushed through from the side where the Bushing is located up to the opposite side of the Clevis Head (8). The Bolt *NAS67103D29* (5) is then tightened by hand with the Castellated Nut *MS9358-016* (6) on the opposite, outer side of the GMA Clevis Head (8). After a hand-tight preload between the Bolt and Nut, the Nut's preload is then increased until the next slot of the nut aligns with the drilled hole of the bolt. This final applied torque is to be controlled and documented. After application of the torque, both parts shall be secured from loosening with the Cotter Pin *MS24665-374*. In accordance to NASA recommendations [RD6], the Cotter Pin shall be bended with pliers as illustrated in Figure 5.

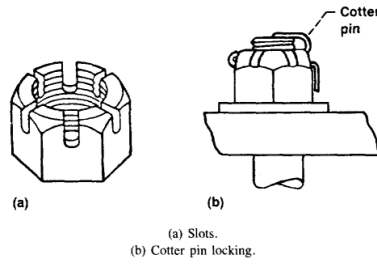


Figure 5: Bending of Cotter Pin [RD6]

## 5. Integration

After the GMA is assembled to a component, it will be integrated to the engine system on one side and the vehicle on the other.

As elaborated in the Definition File [RD1], the GMA is not directly attached to the engine or vehicle. The engine is rather interfaced by a Thrust Dome on the engine side and four Thrust Frame Beams on the vehicle side.

### GMA-Engine

Eight hexagonal head screw fasteners and two  $\varnothing 4m6$  Dowel Pins are provided to connect the GMA with the engine (Figure 6). The GMA is interfaced with the engine by the Thrust Dome [RD1]. The positioning of the Dowel pins are designed for a tighter fit on the Thrust Dome side. Thus, the pins shall be inserted on the Thrust Dome flange first. Thereafter the GMA Lug Head is centered on the Thrust Dome through the Dowel Pins, followed by preloading all eight M6 fasteners in a criss-crossed (“star”) pattern. For a proper execution, the tightening is done in multiple steps, so that all bolts are hand tighten first, then tighten to 30 % of the target torque, followed by 60 % and finally 100 % of the torque listed in the Table 1.

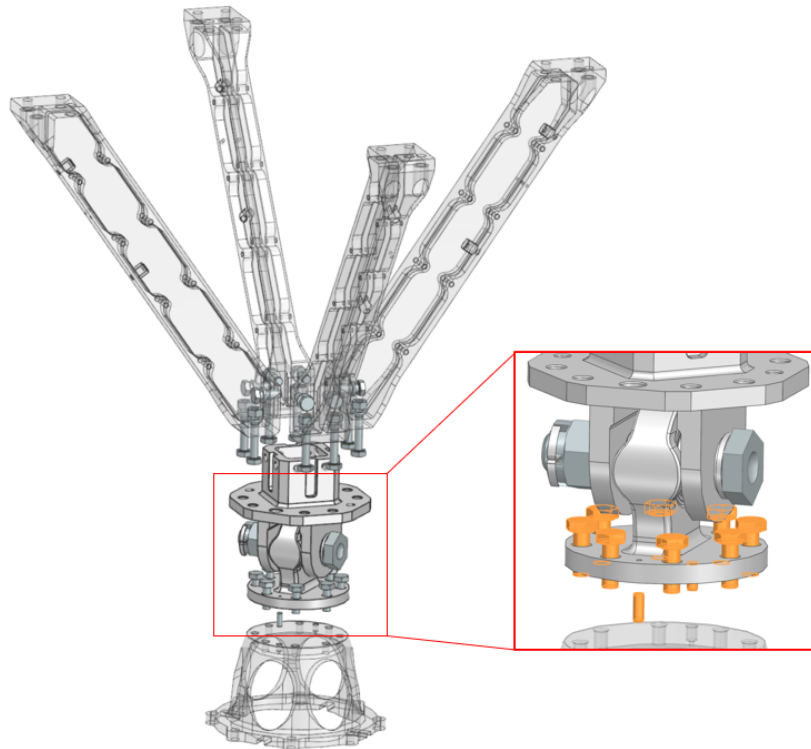


Figure 6: Interface assembly - GMA Lug Head and Thrust Dome

### GMA-Vehicle

As indicated in the Interface Control Document [RD2], 16 fasteners (bolts, washers, nuts) are provided to fix the GMA Clevis Head with the Thrust Frame beams. Figure 7 below shows the configuration and orientation of the fasteners.

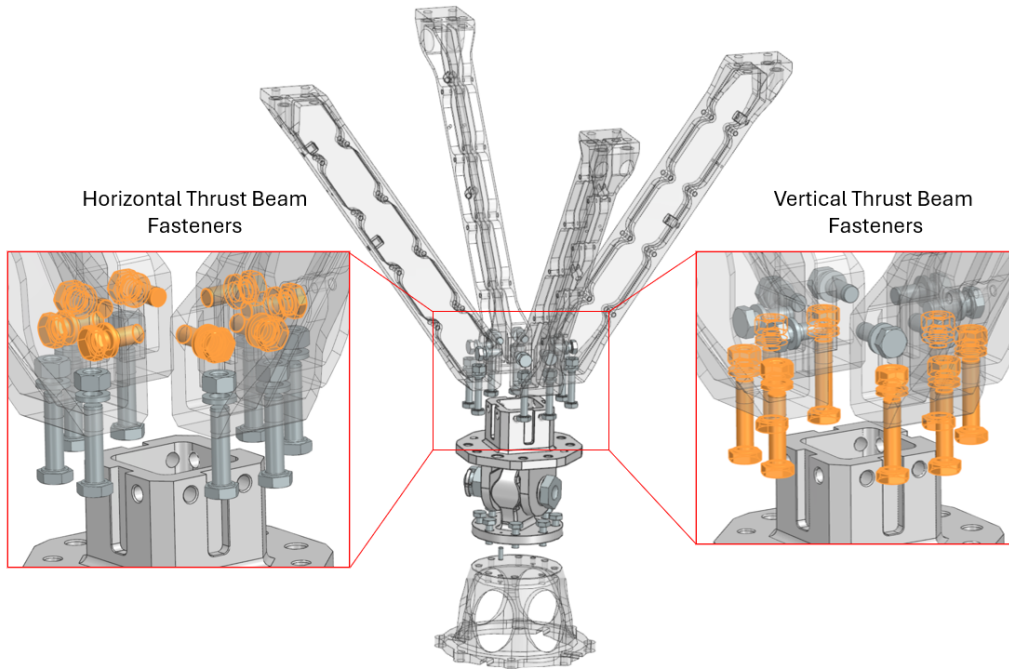


Figure 7: Interface assembly - GMA Clevis Head and Thrust Frame Beams

Before applying torque, all beams shall be appropriately positioned and centered by four centering features as shown in Figure 4 in [RD2]. Since the bolt pattern between the Thrust Beam Structure and GMA Clevis Head is different from standard flange configurations (all bolts in one plane), the torque sequencing of the fasteners might be adapted in further design iteration. In order to provide a homogenous bolt pretension over the entire flange, the preload sequence is recommended to follow the criss-crossed tightening of the vertical fasteners first (x-axis) and then followed by horizontal fasteners in the second step (yz-plane). All four Thrust Frame Beams shall be tightened progressively, not one beam completely before moving to the next. After application of a light seating preload (hand tight) to every single beam, the beams are then preloaded opposite in a criss-crossed pattern as recommended for the GMA Lug Head above in 30 %, 60 % and 100 % of the preload presented in Table 1. Hence, after hand tightening of all fasteners, the preload application shall be sequenced in accordance to the numbering presented in Figure 8 . The image shows a bottom view towards the Thrust Beam Structures in the yz-plane, while the GMA Clevis Head is hidden. The Thrust Vector  $F_{thrust}$  (green) points into the image plane, just like all blue colored fasteners for reference. All red colored fasteners are oriented in the yz-plane (perpendicular to  $F_{thrust}$  and blue colored fasteners).

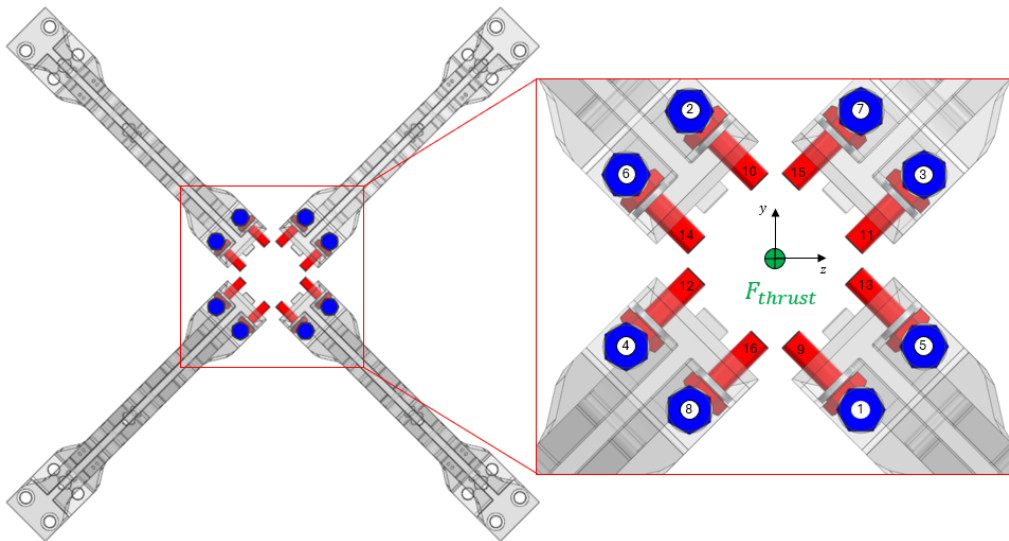


Figure 8: Numbering for preload sequence between GMA Clevis Head and Thrust Frame Beams



Table 1: Preload of fasteners for integration

| Interface                           | Thread type    | Designation         | Preload (Nm) |
|-------------------------------------|----------------|---------------------|--------------|
| GMA Lug Head - Thrust Dome          | Through-Thread | ISO4017-M6x16-A4-70 | ???          |
| GMA Clevis Head - Thrust Frame Beam | Through-Thread | ISO4017-M6x16-A4-70 | ???          |
| GMA Clevis Head - Thrust Frame Beam | Bolt-Nut       | ISO4017-M6x25-A4-70 | ???          |

## 6. Test

All GMA is subjected to a verification programme appropriate to their development status and intended use. In accordance with common ECSS and NASA terminology, the tests are grouped into three main categories:

- Acceptance Tests
- Development Tests
- Qualification Tests

**Acceptance** tests are performed on each manufactured part to verify conformity, workmanship, structural integrity, and basic functionality before assembly to a component and higher-level engine system testing later.

Following successful component acceptance, the hardware proceeds to engine-level testing. **Development** tests support design maturation, characterization, and the resolution of technical issues. **Qualification tests** are performed on a controlled, flight-representative configuration to demonstrate compliance with the specified performance, durability, structural, and safety requirements.

The following chapters describe the objectives, scope, methods, and acceptance criteria for each test category.

### 6.1. Acceptance Test

#### 6.1.1. Inspection

Inspection prior engine operation

Inspection after engine operation

Success criteria

#### 6.1.2. Functional Tests

### 6.2. Development Test

Development tests are conducted during the design-maturation phase to build confidence in new designs and concepts aligned with the Overall Test Plan objectives on the engine system level in accordance to [RD2]. Their objectives are to:

- Characterize the behaviour and performance of the engine;
- Identify weaknesses in the materials, components, interfaces, or overall design;
- Validate and correlate analytical models;
- Investigate operating limits and off-nominal behaviour;
- Support the resolution of design issues; and
- Provide confidence that the final design can successfully complete qualification.

Development-test conditions can extend beyond the normal design or operating range where this is necessary to identify credible weaknesses or design deficiencies. Development testing is iterative and can be performed concurrently with design modifications. However, successful development testing does not, by itself, constitute qualification unless the test article, configuration, test levels, durations, instrumentation, procedures, and success criteria satisfy all applicable qualification requirements.

### **6.3. Qualification Test**

Qualification testing is performed after the design has reached a controlled and sufficiently mature configuration. It is conducted on a dedicated qualification engine or other approved flight-representative engine to demonstrate that the design complies with its specified requirements in the intended operational environments, including the applicable qualification margins. It is assumed that the objectives of the Qualification Tests are defined and executed at engine system level. Hence, the objectives are elaborated in the Huracan Overall Test Plan [RD2]

## **7. AIT documentation**

Assembly and acceptance tests procedures shall be written and executed using a system with version control capability.

Non-conformances shall be documented and tracked in the TEC-internal ERP system and evaluated together with the quality engineers.

## 8. Annex

### 8.1. Instructions: Spherical Bearing Swaging Tool

## 9. Acronym

Table 2: Acronyms

| Acronym | Definition                                    |
|---------|-----------------------------------------------|
| GMA     | Gimbal Mount Assembly                         |
| GSE     | Ground Support Equipment                      |
| MATI    | Manufacturing, Assembly, Integration and Test |

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Instructions for Use of Double Anvil Spherical Bearing Swaging Tool # 10 Wide  
For use with NHBB ABWT-10V and Aurora PWB-10TG, Mil-Spec Part # 14103-10  
The applicable standard governing this process is NAS 0331 Method 100.

- 1) Prepare bore- Bore with Boring head to  $+.0003"/-.0001"$
- 2) Chamfer each side  $.030"/.040"$  by 45 degrees  
Be sure not to deburr the internal angle to bore intersection!
- 3) Press in bearing strait, slowly and carefully  
~~Use Loctite 609 (Green) Bearing and Sleeve Retainer-Do not allow to harden~~
- 4) Bearing should be flush with sides of the housing  $\pm .010"$
- 5) Clean groove in bearing and oil lightly with SAE 30W or equivalent
- 6) Oil the die faces lightly
- 7) Put the arbor through the bearing and put the 60-degree dies (10W-S) on each side of the bearing
- 8) Rotate the dies and feel that they are seated with no foreign objects in the groove
- 9) Using a press apply 21,375 lbs  $\pm$  3,000 (Steel Housings) See Figure 4
- 10) Release press, rotate the dies in the groove
- 11) Re-apply staking load
- 12) Repeat steps 6 to 11 with the 80-degree (10W-F) closing die
- 13) Inspect using a  $.005"$  wire around the staked edge of the bearing- See Figure 5
- 14) A proof load of 1,188 lbs may be applied if desired. This should be applied to the outer race to test the swaging for axial strength.

### NOTES:

A) For a steel housing, begin swaging with a load in the lower part of the swaging pressure tolerance listed above. For an aluminum housing reduce the swaging load to 50% of the value listed above. Increase the load and repeat the procedure until the desired result is obtained, the specifications of Figure 5 must be met. Probe with the sharp corner of a  $.005$  feeler gauge or  $.005$  wire. It is possible to over-stake a bearing and make the ball tight in the assembly (Increase no-load breakaway torque). An ideal process swages the bearing into a housing and the finished swage meets all the Quality Control Criterion of Figure 5 without making the ball tighter in the assembly (increase the no-load breakaway torque).

B) For assemblies to be in full compliance with NAS 0331 Method 100, proof testing (push out test) is required. HPVE manufactures Proof Testing Tools to properly load the bearing without damage. If the QC Criterion of Figure 5 are met, and the procedure above is followed, failure of a proof test is normally not encountered.

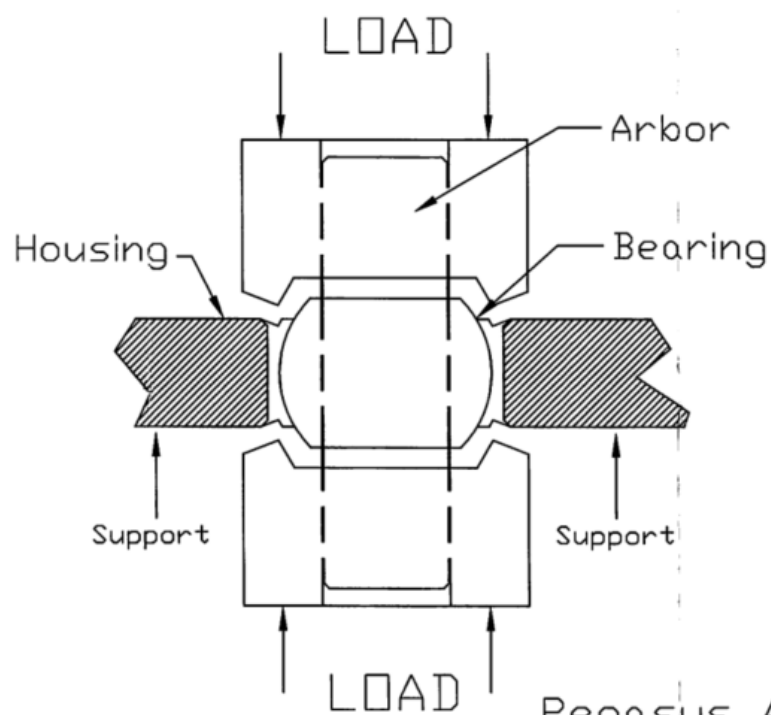
C) In order to remove a swaged bearing for replacement the swaged lip must be cut to allow the lip to fold up when the bearing is pressed out. HPVE manufactures removal tools that allow the technician to remove a swaged in bearing without damage to the housing.

D) Remember: Swaging spherical bearings is an important task, insure that you do it correctly, only you are responsible for your end results.

Thank you for using our product. If you have questions or need assistance feel free to contact us directly. These procedures have been developed over many years and yield excellent results, however technicians should also read and understand NAS 0331 Method 100.

Figure 9: User instructions for swaging

Figure 4 - Anvil Staking Bearings



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Figure 10: Anvil swaging

Figure 5 - Complete Stake Requirements

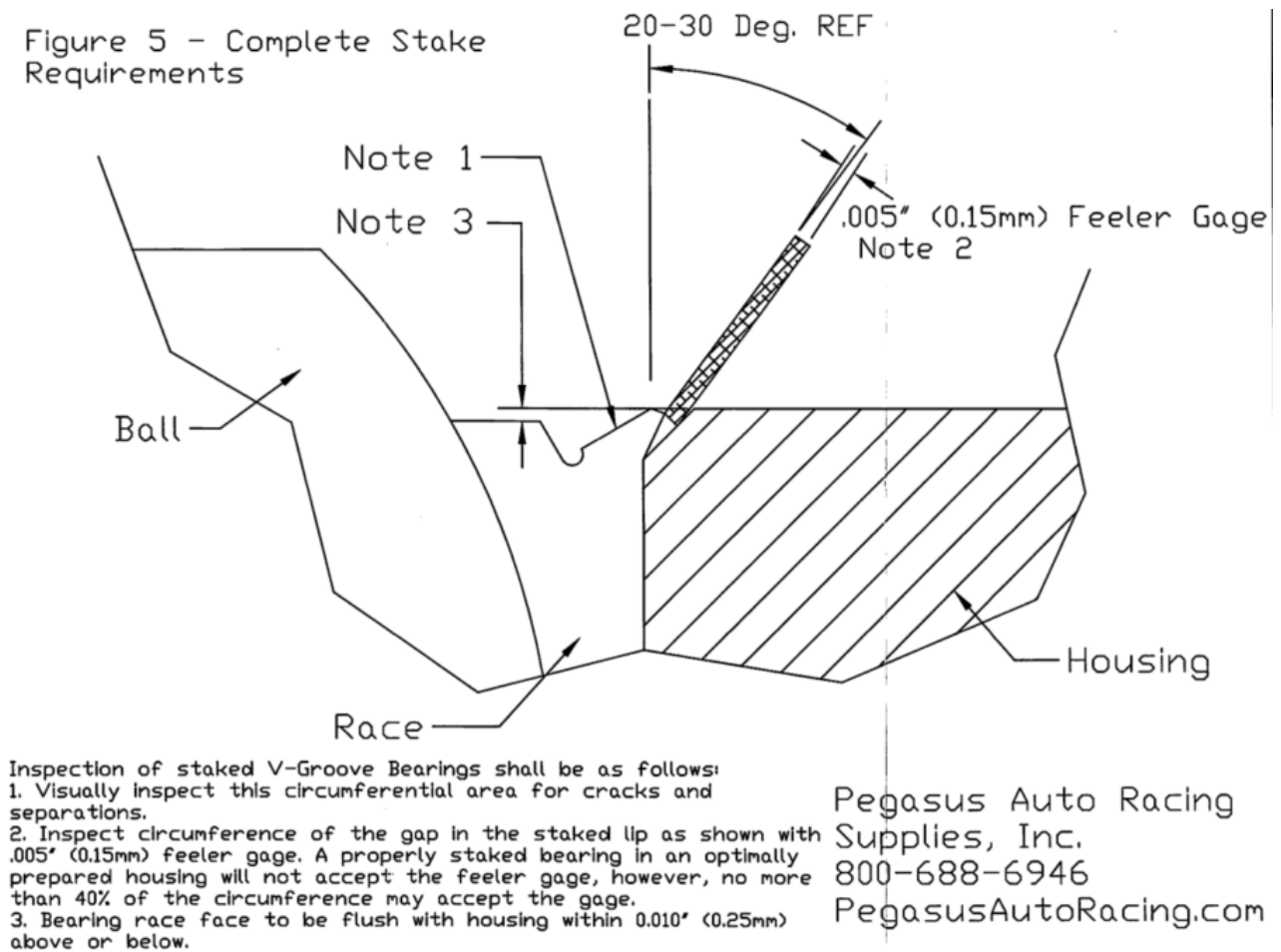


Figure 11: Gauging of swaged bearing